Before Lecture 5

- No matter who you are, i.e., you may be a household or a firm, you will make the following decision considering your opportunity cost:

[1] Quantity demanded (Discussed in Lecture 4)

[2] Quantity supplied

- **Households：** **As the time is limited, you will devote to the job you are good at the most. (Be the labor supplier)**

This topic will be discussed in the course of investment

As the wealth is limited, in addition to consumption, you will make investment and consider how to manage your investment portfolio to maximize your wealth. Buying stocks or bonds? How many? (Be the fund supplier)

- **Firms：** **As the asset and funding are limited, which commodity should be produced to maximize the equity holders' benefits (Be the commodity or service suppliers)**

This topic will be discussed in the course of financial management

Given the product you plan to produce, you will consider the issue of the quantity supplied being able to maximize your equity holders' value and the issue of product pricing.

When facing the two issues, you will have at least two constraints

[1] How many competitors you will face? The demand for the commodity you produce is strong or weak in the short run and long run?

[2] Do you have the production technology that is good enough to making profits? Are your production costs low enough?

Lecture 5 will focus on the constraint [2] and discuss the relation between quantity supplied and cost of production factors

That is, what is the optimal quantity supplied considering the cost of production? What is price of the commodity per unit given the optimal quantity supplied?

Lecture 5: 生產理論與成本分析

(Theory of production and the analysis of cost of production)

Outline

- 5-1: Introduction of firms
- 5-2: Production function and short- and long-run analyses
  - 5-2.1: Short-term analyses for total product, average product and marginal product
  - 5-2.2: Long-run analyses for total product, average product and marginal product
- 5-3: Short- and long-run analyses for costs of production
  - 5-3.1: Short-run analyses for costs of production
  - 5-3.2: Long-run analyses for costs of production
  - 5-3.3: The relation between short- and long-run costs

5-1: 廠商概論

(Introduction of Firms)

- **The Role of a Firm:**
  - Given the product it plans to produce, it is a demander (buyer) in production markets and suppliers of commodity markets

- **Why do firms exist：**
  - Firms have lower transaction costs when making transaction than individuals do
  - For example, instead of contracting a lawyer every time a legal question arises, a firm will hire the lawyer as an employee and form a design team. If the lawyer is a employee inside the firm, their payment is made according to the firm's organization structure. This removes transaction costs that would have encountered in the free market.

- **The goals as firms make decision ：**
  - Given the products individuals or households want to buy
    - Goals when making decision： utility maximization
    - Constraints faced when making decision: limited budget
  - Given the products firms plan to produce
    - Goals when making decision： profits maximization
    - Constraints faced when making decision ：

[1] Asset and funding are limited

∴ production capacity cannot be upgrade to unlimited level

∴ raising funding in capital markets is needed

That will be the topic discussed in the course of financial management.

[2] Limited technology for production

∴ Firms meet production costs.

The more efficient the technology for production, the lower the production costs.

Total Revenue, Total Cost, Profit

- We assume that the firm's goal is to maximize profit.

Profit = Total revenue – Total cost

the amount a firm receives from the sale of its output

TR = P×Q

the market value of the inputs a firm uses in production

Costs: Explicit vs. Implicit

- Two kinds of costs:

  (1) Explicit costs

  - Require an outlay of money
    - ⊙ E.g., paying wages to workers.

  (2) Implicit costs

  - Do not require a cash outlay
    - ⊙ E.g., the opportunity cost of the owner’s time.

- Total cost = Explicit + Implicit costs

Explicit vs. Implicit Costs: An Example

- You need $100,000 to start your business. The interest rate is 5%.
  - Case 1: borrow $100,000
    - ⊙ explicit cost = $5000 interest on loan
  - Case 2: use $40,000 of your savings, borrow the other $60,000
    - ⊙ explicit cost = $3000 (5%) interest on the loan
    - ⊙ implicit cost = $2000 (5%) foregone interest you could have earned on your $40,000.
- In both cases, total (exp + imp) costs are $5000

Economic Profit vs. Accounting Profit

- **Accounting profit**

  = Total revenue - Total explicit costs

- **Economic profit**

  = Total revenue - Total explicit costs - Total implicit costs

Accounting profit ignores implicit costs, so it's higher than economic profit.

- When discussing the decision on consumption, the goal of making this decision is utility maximization and we need to take the consumer’s budget constraint into consideration.

- To analyze the decision on how to maximize profits, we need to take both of the total revenue and total cost of production into account. The total revenue could be related solely to the decision on quantity supplied. Taking it as given, production profits will also be significantly affected by costs of production.

5-2: Production Function and short- and long-run analyses

- To analyze consumers' decisions on quantity demanded, we start from their utility functions. To analyze firms' decisions on quantity supplied, we start form their production functions

- Utility functions describe the relationship between <u>consumption bundles</u> and <u>consumers' satisfaction level</u>
- Production functions describe the relationship between <u>quantity of inputs</u> and <u>quantity of output</u>

- **Production Function**
  - **describing the relationship between quantity of inputs and quantity of output**

- For production, a firm needs use n types of inputs described as $(Y_1, Y_2, \ldots, Y_n)$, where Y_i indicates the quantity of the i-th inputs

- After using $(Y_1, Y_2, \ldots, Y_n)$, the firm gets TP units of output. Then

$$\mathrm{TP} = g(Y_1, Y_2, \ldots, Y_n),$$

where g is the firm's production function.

- Different firms may need different production inputs and thus may have different forms of production functions

- For ease of discussion, the inputs are simplified from n types to 2 types (Y_1, Y_2). Let $Y_1 \equiv L$, indicating the quantity of labor input. Let $Y_2 \equiv K$, indicating the quantity of capital inputs, such as equipment and factory building.

Thus, the production function is simplified as $\mathrm{TP} = g(L, K)$;

e.g., $\mathrm{TP} = 5 \cdot \mathrm{K} \cdot \sqrt{\mathrm{L}}$

- **Decision on short- and long-run production**

  - A firm will make short-run (e.g., the plan within 1 month) and long-run (e.g., the plan within 20 years) decisions

  - For example, a firm plans to expand its production capacity. It can in the short-run ask its employee to work overtime or hire new employees in the labor market. In the long-run, it can renew the equipment of its production lines

- For ease of analysis, the production period could be identified as short-run (SR) and long-run.

  [1] In the short-run, the quantity of input being able to be changed and adjusted is called "Variable Factors."
  Those cannot be adjusted is "Fixed Factors."

  [2] Both of the quantity of variable and fixed factors can be adjusted in the long-run.

- Thus, for the analysis of short-run production, we take K as given. That is, we let $K = K_0$ and let the production function be the form of

$$\textbf{S}\mathrm{TP} = g(L, K_0).$$

This implies that in the **Short** run we discuss how the change in quantity of variable factors affects TP by taking the quantity of fixed factors as given.

5-2.1: 總產量、平均產量、邊際產量的短期分析

(Short-run analyses for total product, average product and marginal product)

- **Short-Run Total Product (STP): We take $K = K_0$ as given. The relation between variable factors, e.g., the labor, and TP is**

$$\mathbf{STP} = g(L, K_0)$$

- **Short-Run Average Product (AP):**

  The total product divided by quantity of variable factor, e.g., the labor (a worker), is $\mathbf{SAP_L}$,

$$\mathbf{SAP_L} = \frac{STP}{L} = \frac{g(L,K_0)}{L}$$

- **Short-Run Marginal Product (SMP):**

  We take $K = K_0$ as given and let the variable factor be the labor input. Thus the marginal product of labor is

$$\mathbf{SMP_L} = \frac{\Delta STP}{\Delta L} = \frac{g(L+\Delta L,K_0)-g(L,K_0)}{\Delta L}$$

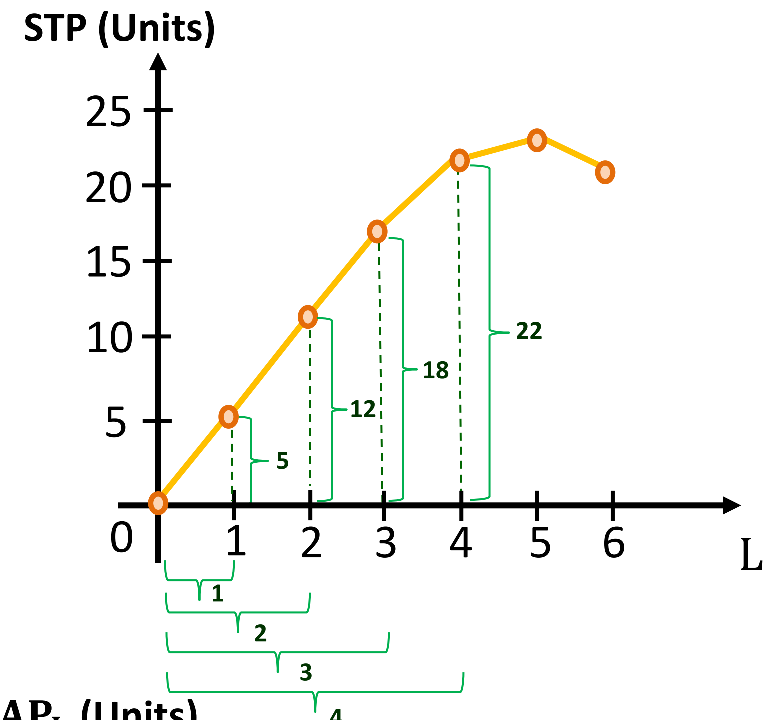
- 短期總產量、平均產量、邊際產量三者的計算

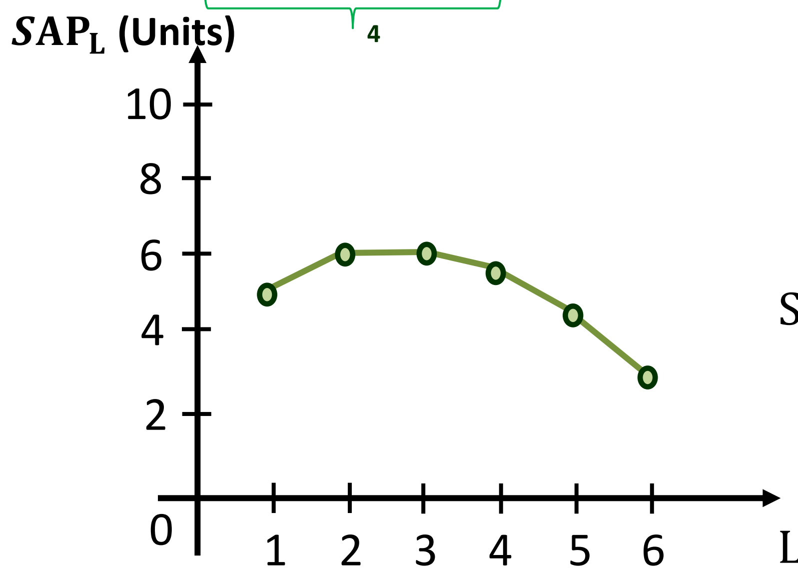
(The calculation of STP, SAP_L and SMP_L)

	L	STP	SAP_L
A	0	0	-
B	1	5	5.0
C	2	12	6.0
D	3	18	6.0
E	4	22	5.5
F	5	23	4.6
G	6	20	3.3

(這個表可以想像成，一個勞工，L是他投入工作的時數，L=1代表工作1個小時)

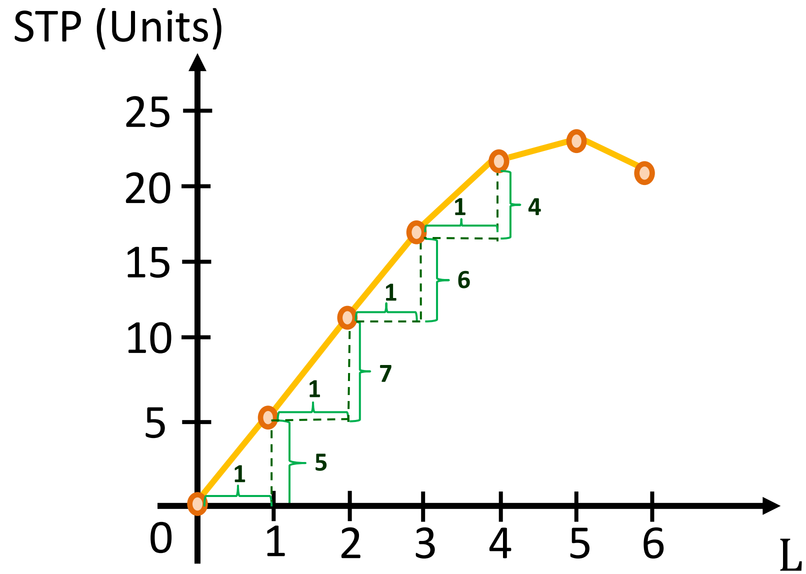
(L = 1 could be a worker’s 1-hour dedication to production)

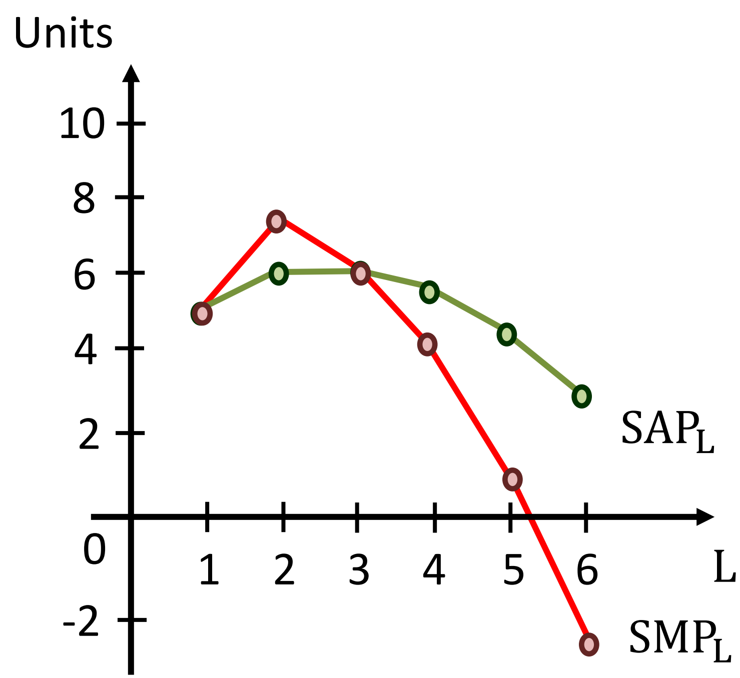




$$SAP_L = \frac{STP}{L}$$

	L	STP	SAP_L	SMP_L
A	0	0	-	
B	1	5	5.0	5
C	2	12	6.0	7
D	3	18	6.0	6
E	4	22	5.5	4
F	5	23	4.6	1
G	6	20	3.3	-3



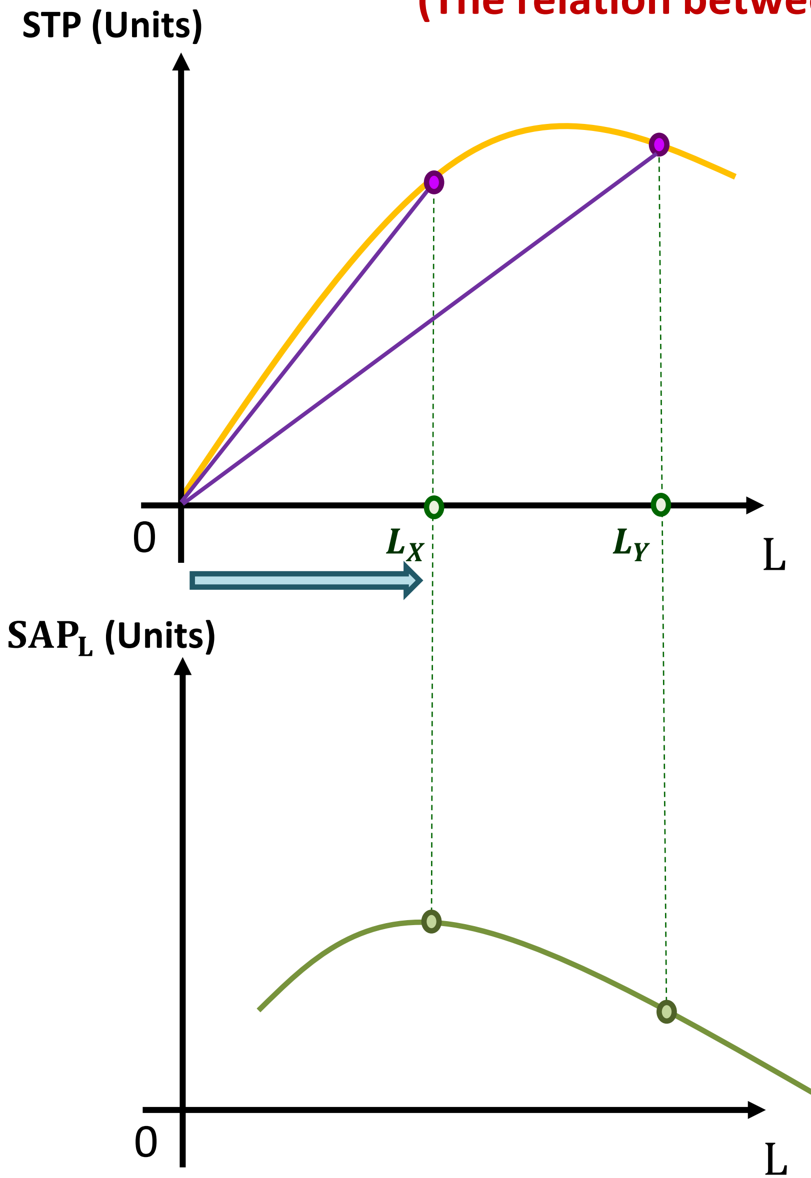


$$SAP_L = \frac{STP}{L}$$

$$SMP_L = \frac{\Delta STP}{\Delta L}$$

- # 短期總產量、平均產量兩者之間的關係

(The relation between STP and SAP$_L$)



- **STP線上任何一點與原點連線的斜率為該點對應的SAP$_L$**

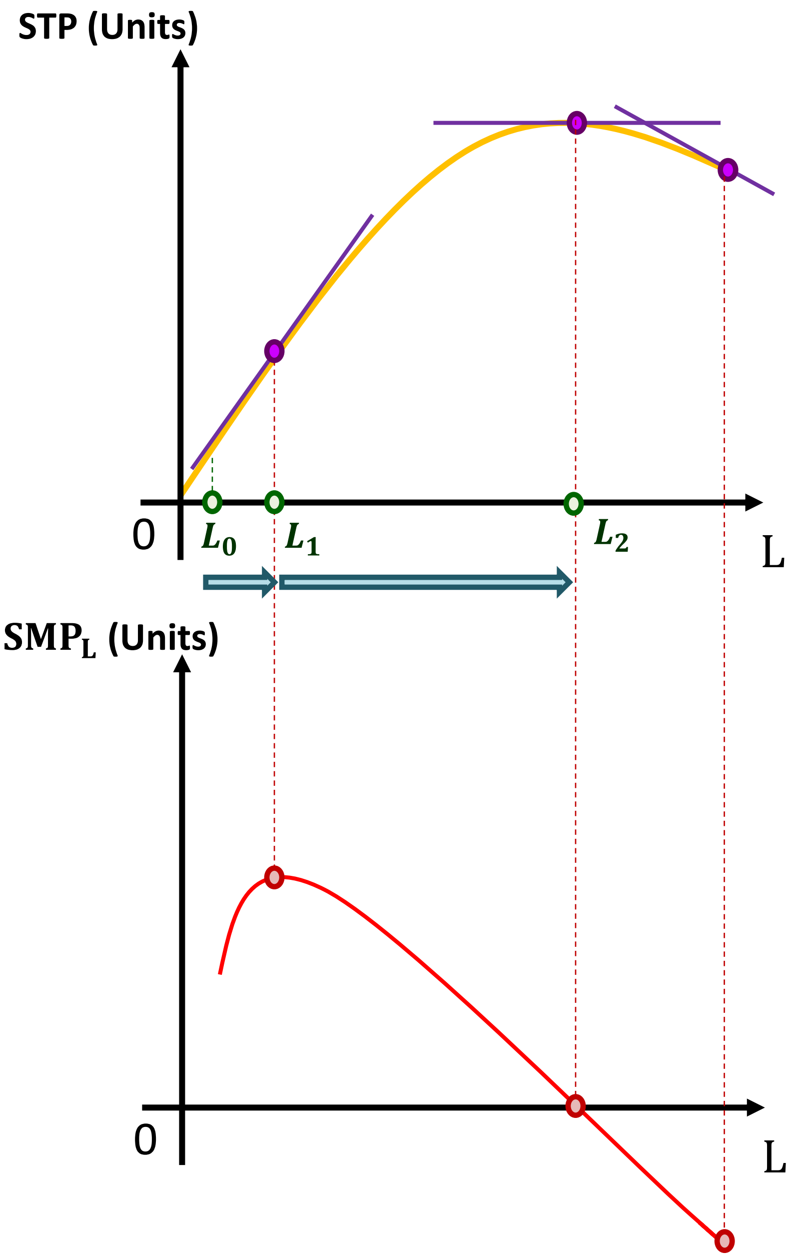
Given that we have STP $= g(L, K_0)$. SAP$_L$ is slope of the line between the point on the TP curve and origin.

- **STP線與原點的連線的斜率在L$_X$處達到最大，所以SAP$_L$在此處達到最高，之後開始遞減**

The SAP$_L$ curve has maximum when $L = L_x$.

The SAP$_L$ curve is usually a inverted U-shape.

- **The relation between STP and SMP_L**



- **The tangent slope of STP curve is SMP_L**

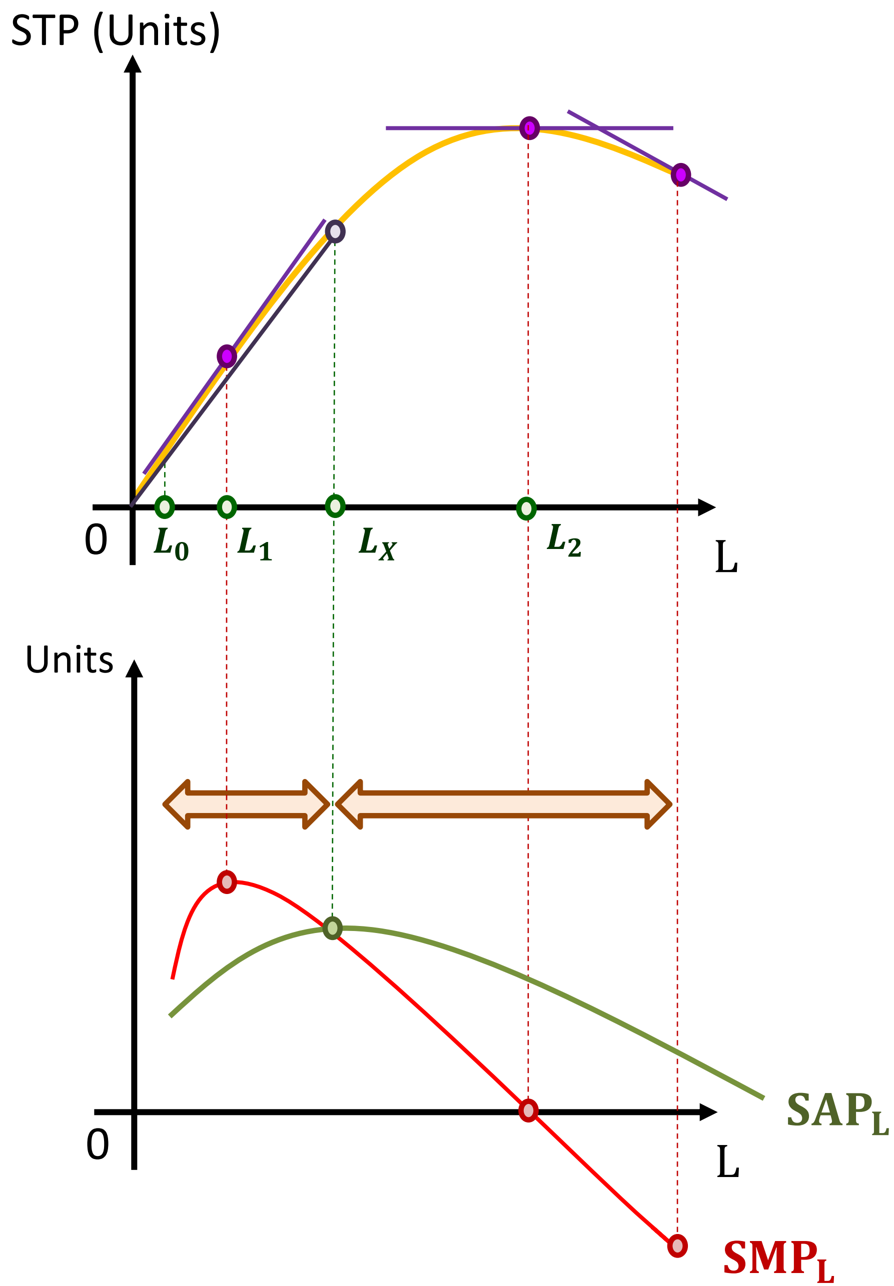
- **If the STP curve is inverted U shape, the SMP_L is also inverted U shape.**

- **That is, SMP_L has maximum. Unlike SAP_L, SMP_L could be positive, zero, or negative.**

- **Inverted U shape of SMP_L implies that the marginal product of an input declines as the quantity of the input increases.**

  This property is called Law of diminishing marginal returns

- **The relation between $\mathrm{SAP_L}$ and $\mathrm{SMP_L}$**



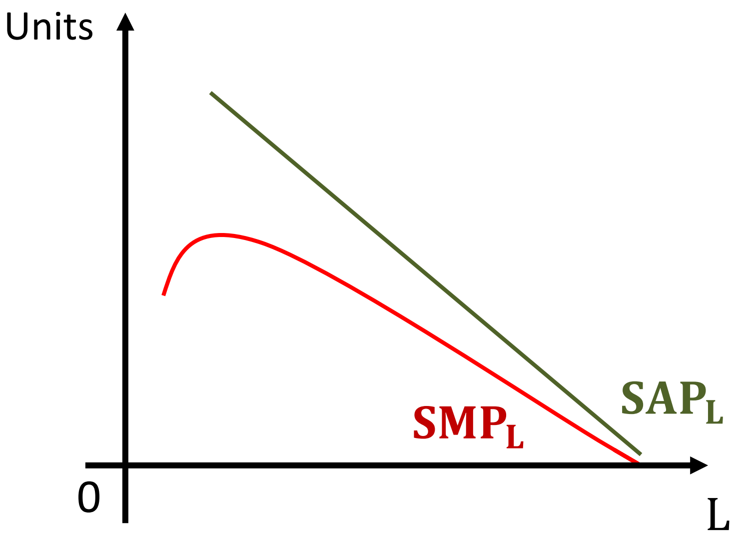
- **When $\mathrm{SAP_L} < \mathrm{SMP_L}$, the $\mathrm{SAP_L}$ curve is positively-sloping**

- **When $\mathrm{SAP_L} > \mathrm{SMP_L}$, the $\mathrm{SAP_L}$ curve is negatively-sloping**

- **When $\mathrm{SAP_L} = \mathrm{SMP_L}$, the $\mathrm{SAP_L}$ has maximum. That is, $\mathrm{SAP_L}$ and $\mathrm{SMP_L}$ curves intersect at the point where $\mathrm{SAP_L}$ has maximum.**

- **Case 1: $SAP_L > SMP_L$ for all L**

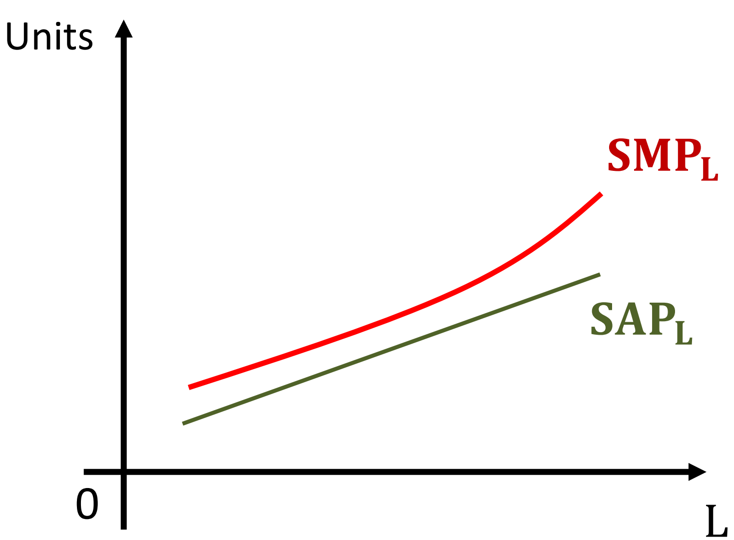
E_Lecture5_W1_Q1-Q2



- 當邊際產能皆小於平均產能，則平均產能會隨勞動力增加而遞減

(Then SAP_L curve is downward- sloping for all L)

- **Case 2: $SAP_L < SMP_L$ for all L**



- 當邊際產能皆大於平均產能，則平均產能會隨勞動力增加而遞增

(Then SAP_L curve is upward- sloping for all L)

- 但如果邊際產能符合邊際報酬遞減法則，則 SMP_L 線會變成倒U形，左圖會變成與上頁投影片一樣

(If SMP_L follows diminishing marginal product, then SMP_L curve is inverted U shape. Once SMP_L curve is inverted U shape, SAP_L has maximum.)

5-2.2: 總產量、平均產量、邊際產量的長期分析

(Long-run analyses for total product, average product and marginal product)

- **Long-Run Total Product： In the long run, the firm can adjust the quantity of fixed and variable factors. Thus,**

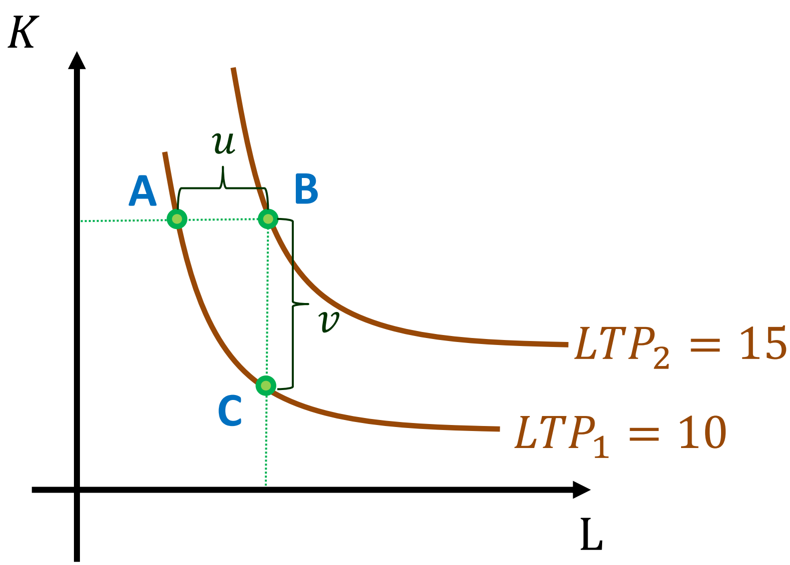
$$\mathbf{LTP} = g(L, K).$$

Given an input bundle (L, K), we obtain TP through a production function

- **Isoquant Curve：**

The curve on the L-K plane that indicates a specific LTP generated by different input bundles.

- **Marginal Rate of Technical Substitution (MRTS):**

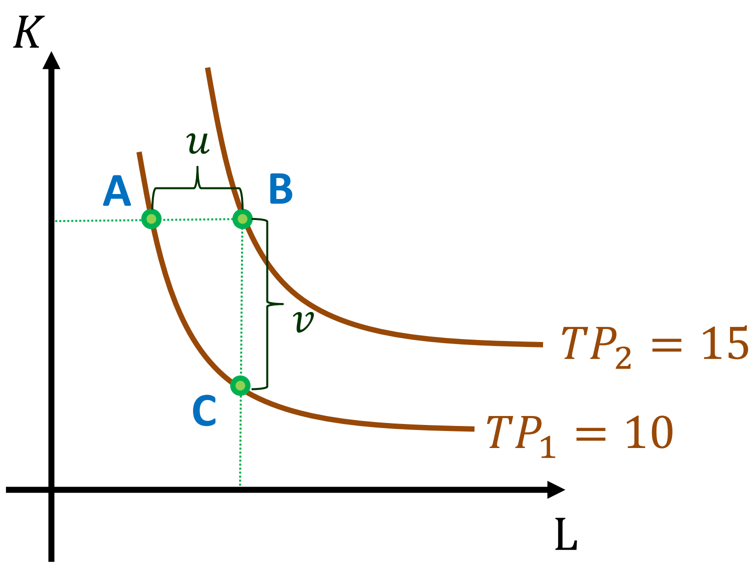


- In the left figure, we have two isoquant curves indicating $LTP_1 = 10$ and $LTP_2 = 15$. If increase L by ΔL, $\Delta L = u$ (from point A to B), then ΔLTP = LTP_2 - LTP_1= 5. Thus, we have marginal product of L

$$LMP_L = \frac{\Delta LTP}{\Delta L} = \frac{5}{u}$$

- If we decrease K by ΔK, $\Delta K = v$ (from point B to C), then we also have ΔLTP = LTP_2 - LTP_1= 5. Thus, we have marginal product of K

$$LMP_K = \frac{\Delta LTP}{\Delta K} = \frac{-5}{v}$$

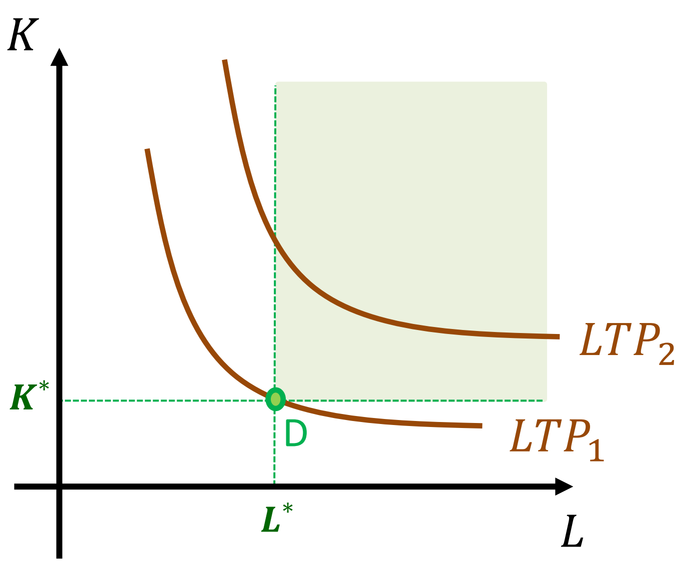


- The MRTS at point A could be described through marginal product of L and K by

$$\mathrm{MRTS_A} = \frac{v}{u} = \left|\frac{5/u}{-5/v}\right| = \left|\frac{\Delta LTP/\Delta L}{\Delta LTP/\Delta K}\right| = \left|\frac{LMP_L}{LMP_K}\right|$$

- MRTS implies that if a producer want to keep its LTP constant, the decrement in one production factor needs to be substituted for the increment in the other production factor

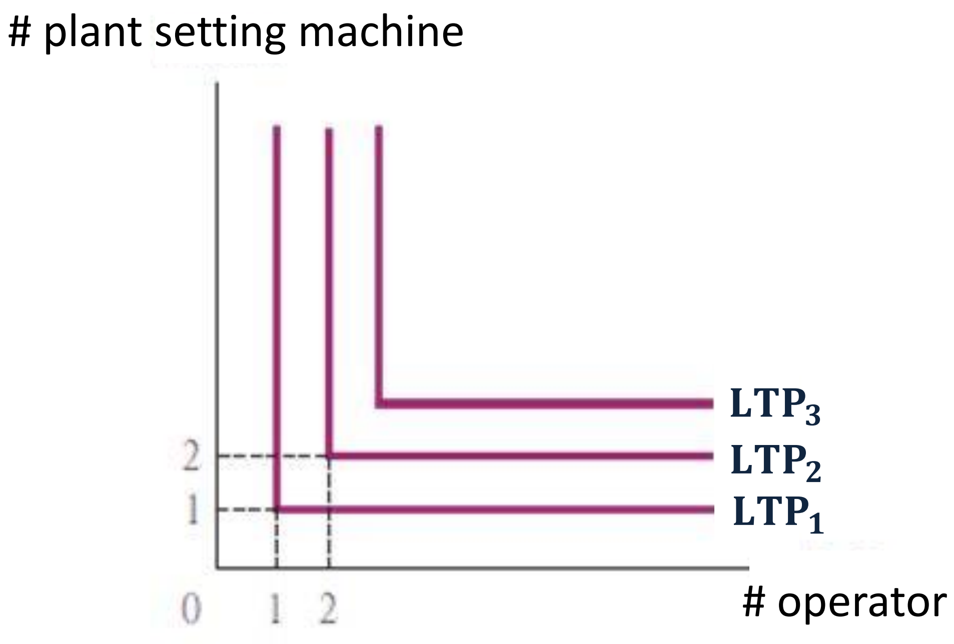
- If long-run marginal product (LMP_L or LMP_K) follows the law of diminishing marginal returns, the MRTS is also diminishing. That is, to keep LTP constant, the input you need to remove to compensate for an increase in another input will increase once the quantity of former inputs is already high.



- As the long-run marginal product is positive, the isoquant curves in the first quadrant indicate that more inputs used for production. That is, more inputs are used to produce LTP_2 than those used to produce LTP_1.
- Due to the fact that MRTS is diminishing, the isoquant curves are bowed inward.
- Isoquant curves cannot cross.

- **Two special isoquant curves**

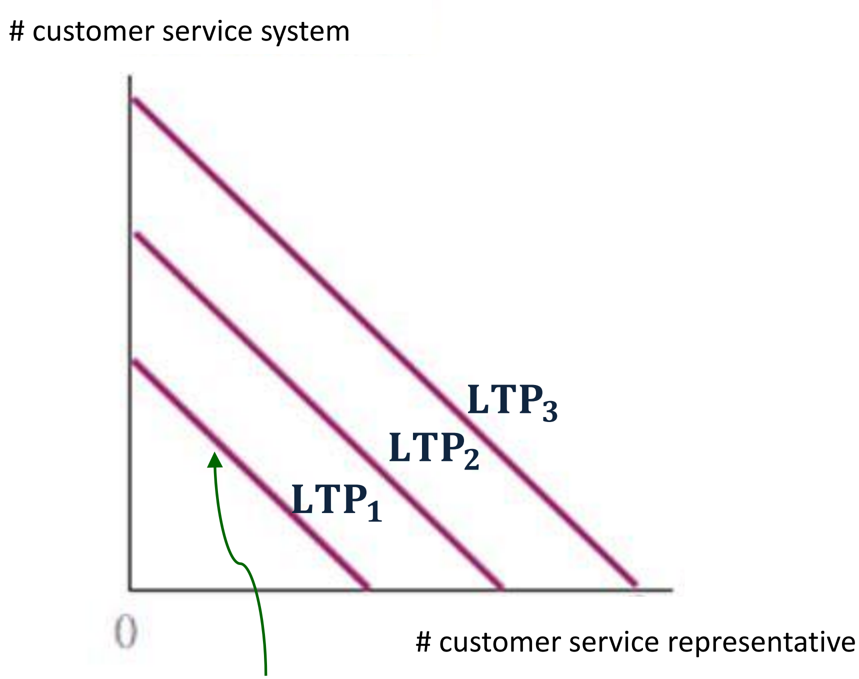
  [1] two inputs are perfect complements



- **In terms of plant cultivation, plant setting machines and operators are two kinds of inputs. Note that one machine need to be operated by one operator (like the left figure). In the case, we state that the two inputs are perfect complements.**

- **When production is needed to be made through some specific combination of input, those inputs are perfect complements for that production.**

[2] two inputs are perfect substitutes



The slope of the isoquant curves are -1

- In terms of customer service, customer service system and representative are perfect substitutes

- For this case, the slope of the isoquant curves is -1 for all. If the firm has budget constraint, it will choose the cheaper inputs.

E_Lecture5_W1_Q3

5-3: The analyses of short- and long-run production costs

- Suppose that only two inputs are needed: labor (variable factor) and capital such as equipment (fixed variable)

- Once a worker is hired, the producer needs pay salary w for the use of per unit of labor. w is called wage rate.

- Once the equipment is rented, the producer needs pay rental r for the use of per unit of equipment.

5-3.1: 成本的短期分析

(Short-run analyses for costs of production)

- 在短生產期間內，只有變動生產要素可以調整，例如勞動力

  (In the short run, producers can only adjust the quantity of variable factor inputs, such as labor)

- **Short-run Total Cost (STC): The market value of the inputs a firm uses in production. It includes**

  [1] Short-run Total Fixed Costs (STFC): Costs that do not vary with the quantity of output produced

  [2] Short-run Total Variable Costs (STVC): Costs that vary with the quantity of output produced

- **Thus, STC = STFC + STVC**

- **Given STC, STFC, and STVC, we can then define short-run average cost (SAC), short-run average fixed cost (SAFC), and short-run average variable cost (SAVC):**

Given short-run total product STP,

then SAC is $\mathbf{SAC = \frac{STC}{STP}}$

SAFC is $\mathbf{SAFC = \frac{STFC}{STP}}$

SAVC is $\mathbf{SAVC = \frac{STVC}{STP}}$

- **Thus** $\mathbf{SAC = \frac{STC}{STP} = \frac{STFC+STVC}{STP}} =$ **SAFC + SAVC**

- **With TC, we can define short-run marginal cost (SMC):**

  the increase in total cost that arises from an extra unit of production

$$\mathbf{SMC} = \frac{\Delta \mathbf{TC}}{\Delta \mathbf{TP}}$$

- **Notice that TFC will not increase with TP in the short run**

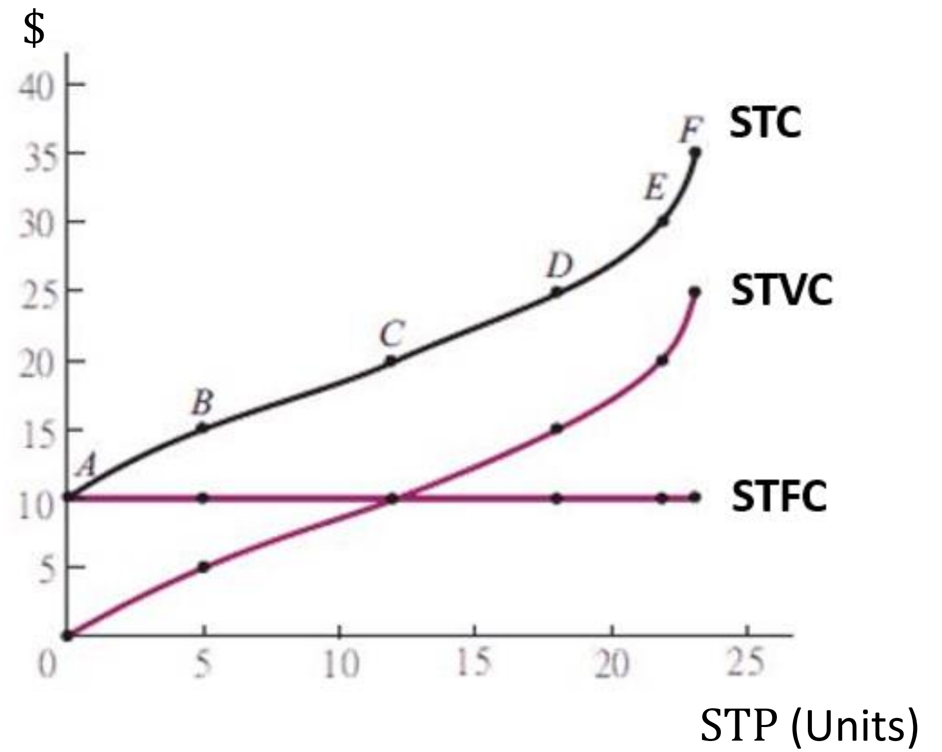
  Thus,

$$\mathbf{SMC} = \frac{\Delta \mathbf{STC}}{\Delta \mathbf{STP}} = \frac{\Delta \mathbf{STFC} + \Delta \mathbf{STVC}}{\Delta \mathbf{STP}} = \frac{\Delta \mathbf{STVC}}{\Delta \mathbf{TP}}$$

- 我們先有短期總產量(STP)、短期總成本(STC) 、短期總固定成本(STFC)、短期總變動成本(STVC)

(We take STP, STC, STFC, and STVC as given.)

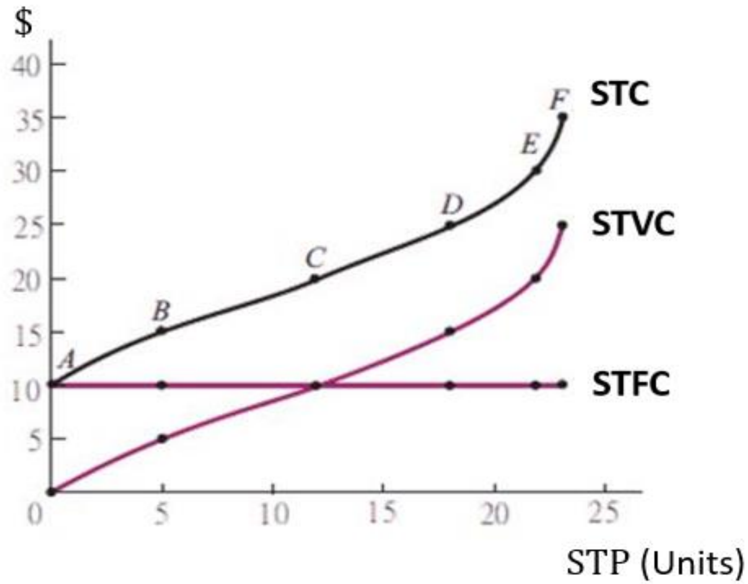
	L	STP	STC	STFC	STVC
A	0	0	10	10	0
B	1	5	15	10	5
C	2	12	20	10	10
D	3	18	25	10	15
E	4	22	30	10	20
F	5	23	35	10	25

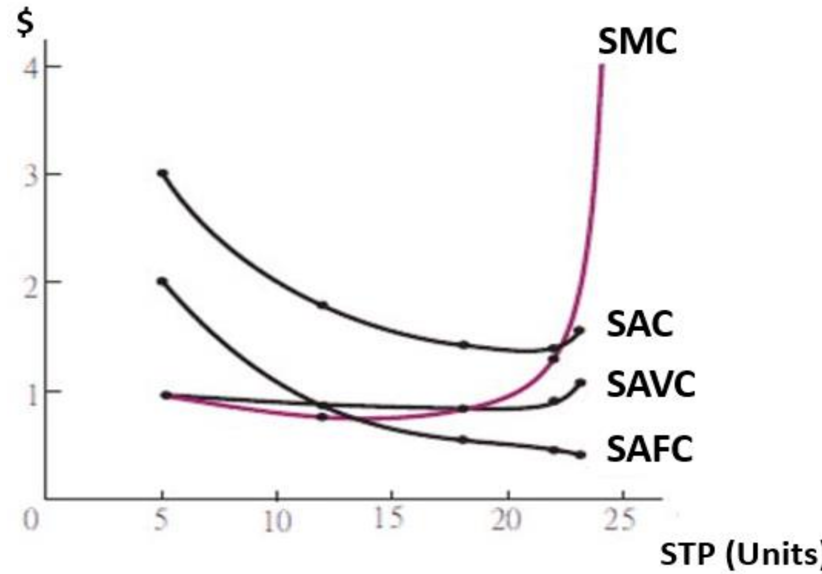


- 短期平均成本(SAC) 、短期平均固定成本(SAFC)、短期平均變動成本(SAVC) 、短期邊際成本(SMC) 四者的計算

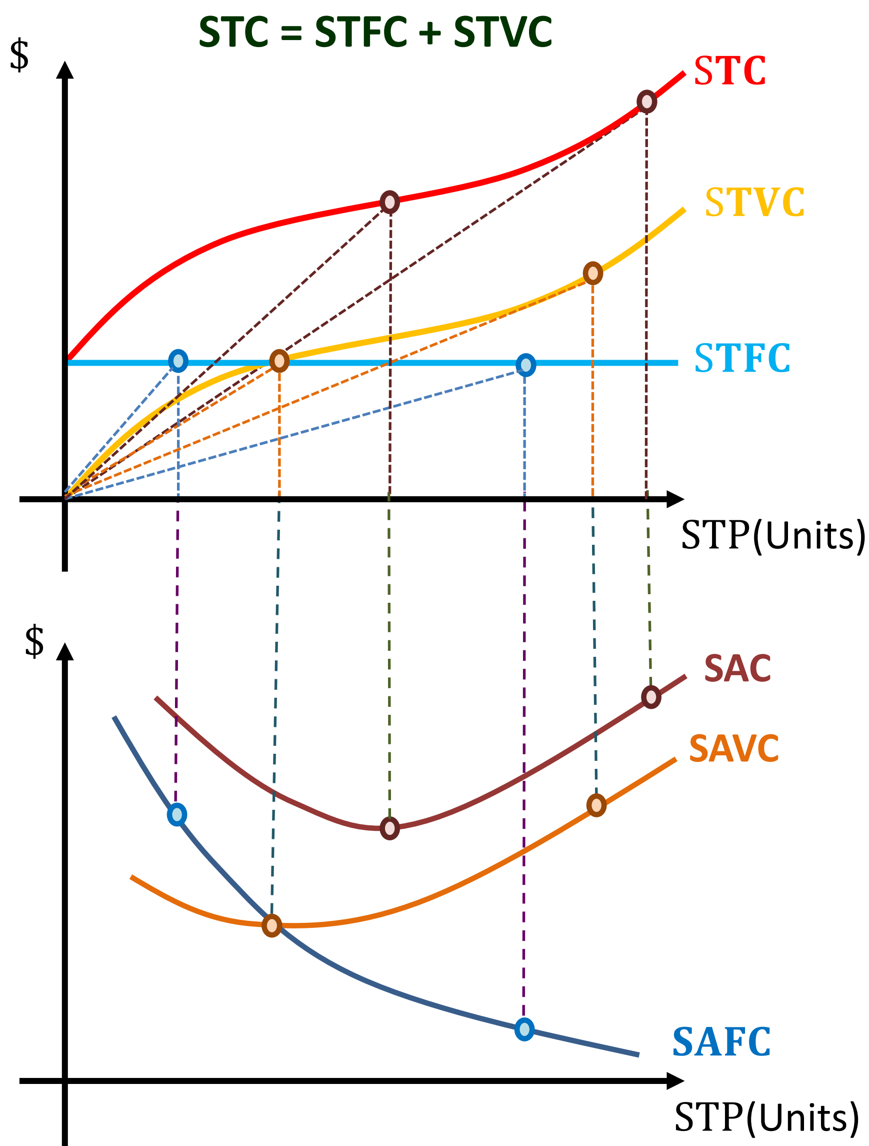
(Then we calculate SAC, AFC, AVC, and SMC)

	L	STP	STC	STFC	STVC	SAC	SAFC	SAVC	SMC
A	0	0	10	10	0	-	-	-	-
B	1	5	15	10	5	3.00	2.00	1.00	1.00
C	2	12	20	10	10	1.67	0.83	0.83	0.71
D	3	18	25	10	15	1.39	0.56	0.83	0.83
E	4	22	30	10	20	1.36	0.45	0.91	1.25
F	5	23	35	10	25	1.52	0.43	1.09	5.00
						STC/STP	STFC/STP	STVC/STP	ΔSTVC/ΔSTP





- The relation among STC, STFC, STVC, SAC, SAFC and SAVC.



- The values of the points on SAFC, SAVC, and SAC are the slope of the lines connected by the origin and the points on STFC, STVC, and TC because according to the definition

$$\mathrm{SAFC} = \frac{\mathrm{STFC}}{\mathrm{STP}} \text{、} \mathrm{SAVC} = \frac{\mathrm{STVC}}{\mathrm{STP}} \text{、}$$

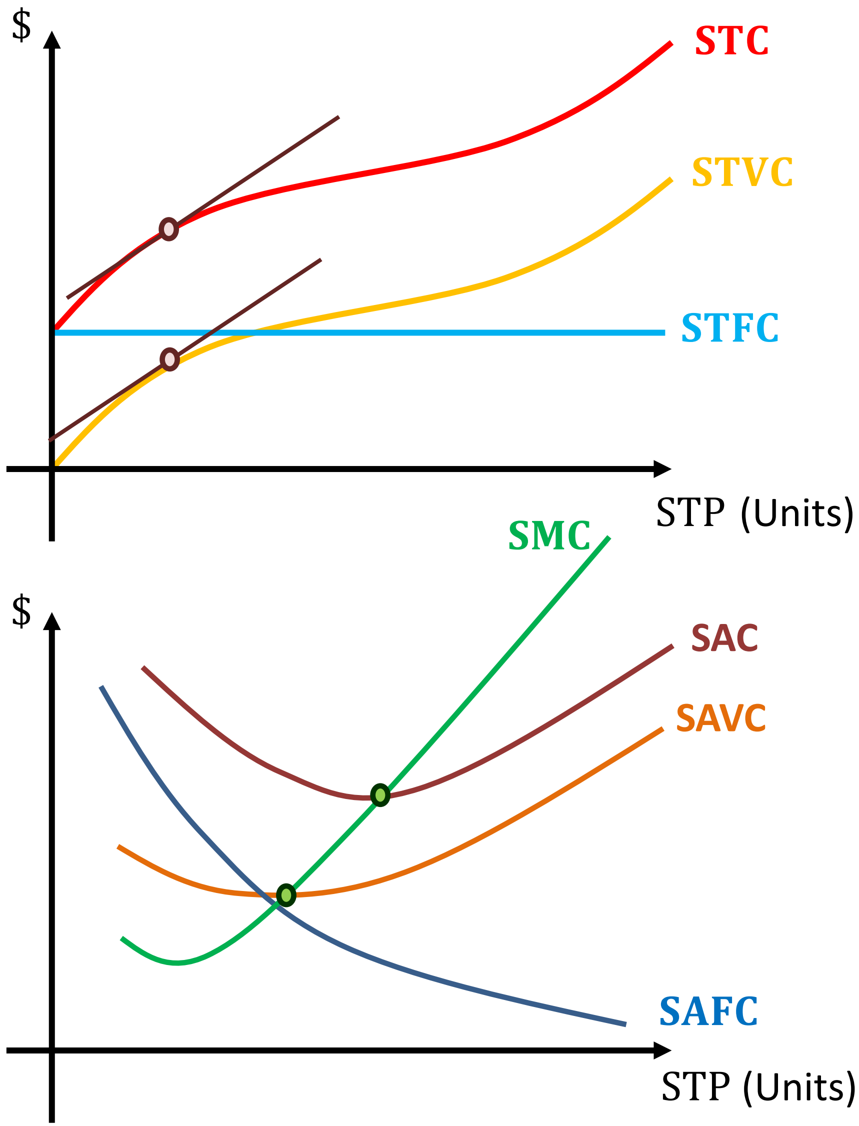
$$\mathrm{SAC} = \frac{\mathrm{STC}}{\mathrm{STP}} = \frac{\mathrm{STFC+STVC}}{\mathrm{STP}} = \mathrm{SAFC} + \mathrm{SAVC}$$

- SAFC decreases as STP increases because STFC curve is horizontal

- SAC curve is U-shaped. That is, SAC is first decreasing and then increasing as STP increases. The decreasing pattern is due to the fact that the SAFC curve is decreasing

- **The relation among SAC, SAVC, and SMC**

E_Lecture5_W2_Q1-Q2



- **SMC is the slope of the tangents of the TC curve, because**

$$\mathrm{SMC} = \frac{\Delta \mathrm{STC}}{\Delta \mathrm{STP}} = \frac{\Delta \mathrm{STVC}}{\Delta \mathrm{STP}}$$

- **As SMC < SAC, SAC is decreasing.**
  As SMC = SAC, SAC has the minimal value;
  As SMC > SAC, SAC is increasing

- **The relation between SMC and SAVC is the same with that between SMC and SAC**

- **Thus the SMC curve will cross the lowest points of SAC and SAVC curves**

- **SAC curve is U-shaped. SAC is first decreasing because the SAFC curve is decreasing. SAC then increasing because the SMC curve is increasing**

5-3.2: 成本的長期分析–
使用等成本線、等產量線分析法

(Long-run analyses for costs of production – using isocost and isoquant curves)

- 在長生產期間內，無論是資本…等固定生產要素，例如廠房的租賃，或勞動力…等變動生產要素，都可以調整

  (In the long run, producers can adjust the quantity of variable and fixed factor inputs, such as labor and production equipment)

- 假定每單位變動生產要素的使用價格稱為w，每單位固定要素的租賃成本為r

  (Suppose that the cost of using per unit of variable factor is w and the cost of using per unit of fixed factor is r)

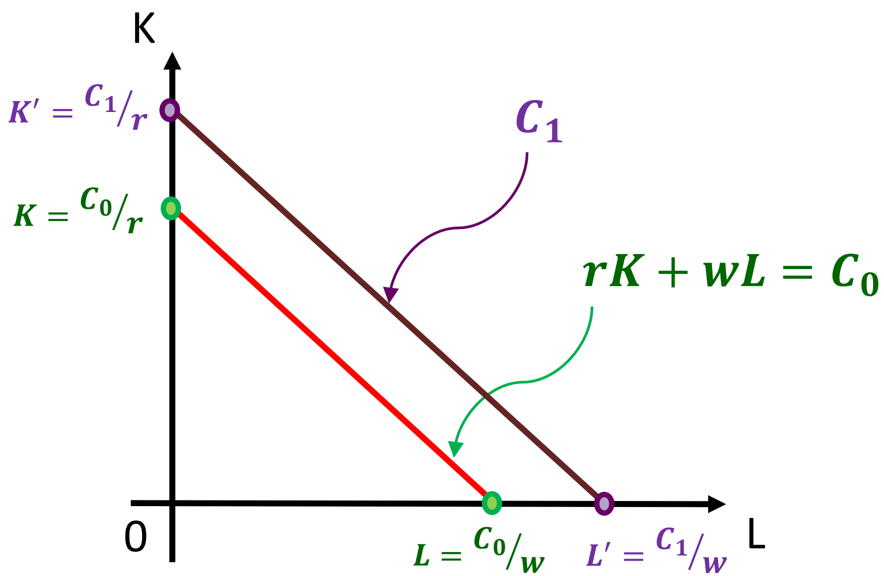
- **Isocost Curve:**

Suppose that a firm only use two kinds of inputs, one is L units of labor, the variable factors, and the second is K units of equipment, the fixed factors. Thus, the firm's total product in the long run is

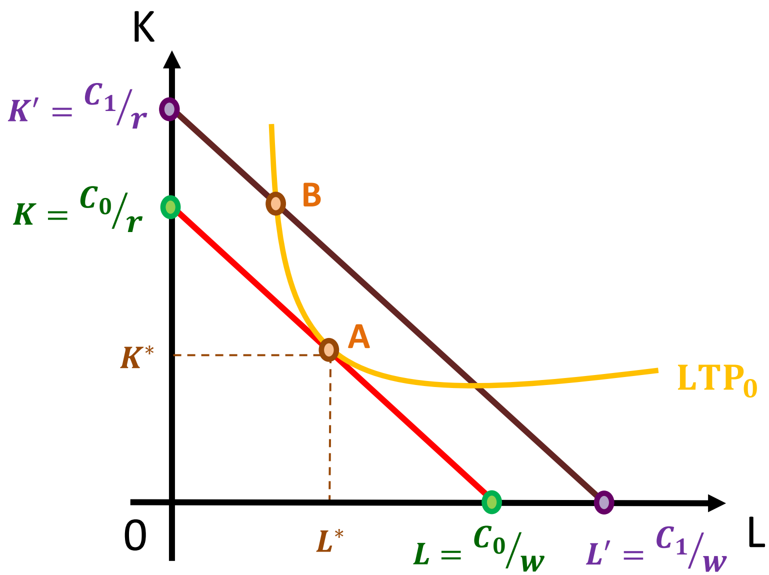
$$\mathbf{LTP} = g(L, K)$$

If the cost of using one unit of equipment is w and the cost of using one unit of labor is r, then the firm's total cost is $\boldsymbol{C_0}$ and the isocost curve can be described as

$$\boldsymbol{C_0} = rK + wL$$



- **In the left figure, $\boldsymbol{C_1 > C_0}$**
- **The slope of the two isocost curves is $\boldsymbol{\frac{w}{r}}$**



- **Given that the level of total product is $\mathbf{LTP_0}$ displayed through the yellow isoquant curve, the lowest cost could be the A on the isocost curve C_0 instead of B on the curve C_1**

- **A indicates the input bundle (L^*, K^*)**

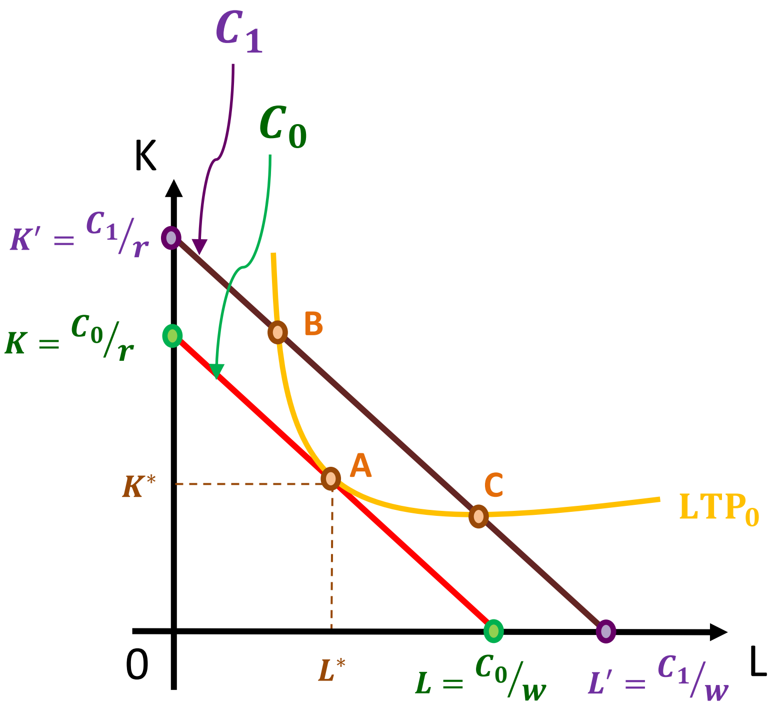
- **The MRS at point A on the isoquant curve $\mathbf{TP_0}$ is**

$$\mathrm{MRTS_A} = \left|\frac{MP_L^A}{MP_K^A}\right|;$$

the slope of isocost curve C_0 is $\frac{w}{r}$.

- **The tangent point A here means**

$$\mathrm{MRTS_A} = \left|\frac{MP_L^A}{MP_K^A}\right| = \frac{w}{r}$$



- Given that the level of total product is $\mathrm{TP_0}$, point A is the corresponding lowest cost C_0 with input bundle (L^*, K^*), and

$$\mathrm{MRTS_A} = \left|\frac{MP_L^A}{MP_K^A}\right| = \frac{w}{r}$$

- Given $\mathrm{LTP_0}$, then the point B implies

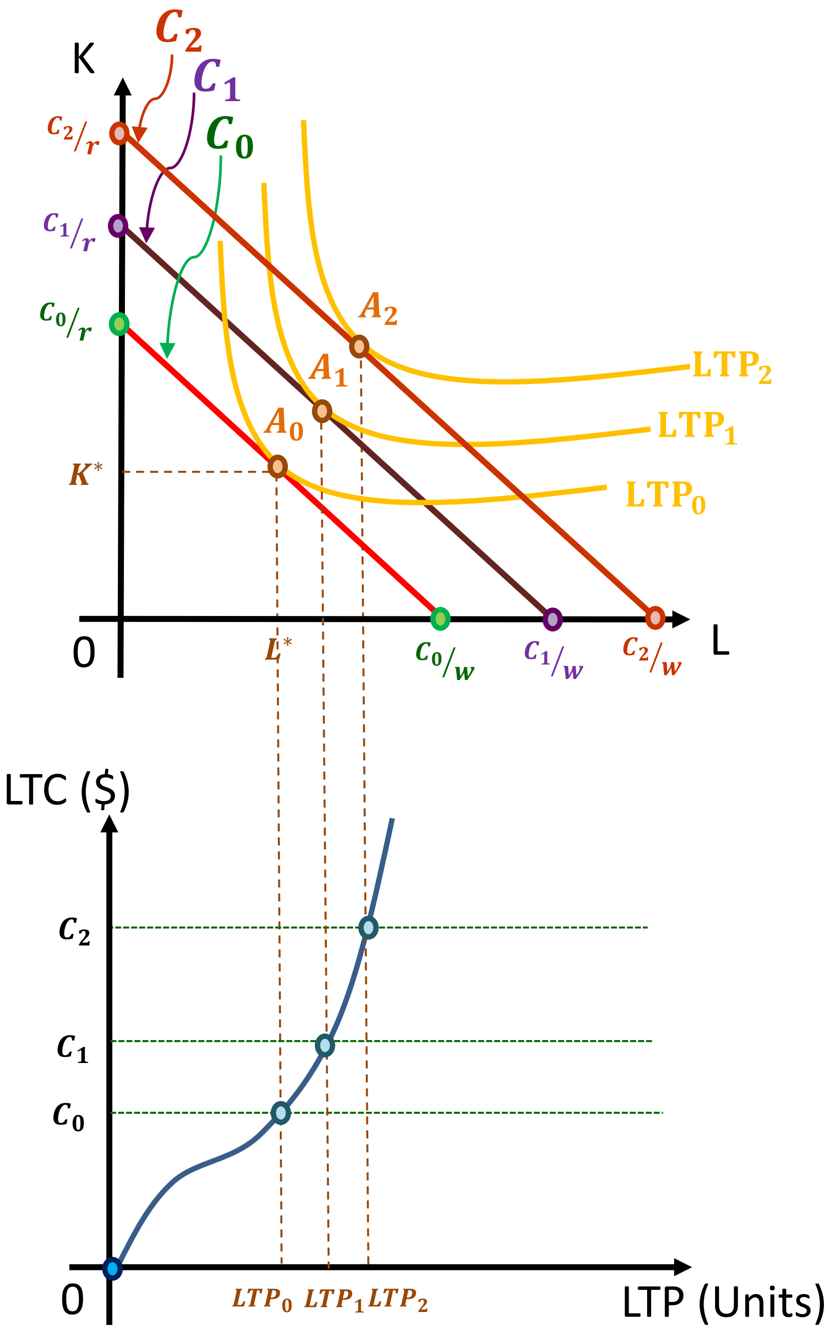
$$\mathrm{MRTS}_B = \left|\frac{MP_L^B}{MP_K^B}\right| > \frac{w}{r}, \text{ and}$$

total cost $C_1 > C_0$. The firm can try to decrease the usage of equipment but increase the usage of labor to make its total production cost lower

- Given $\mathrm{LTP_0}$, then the point C implies

$$\mathrm{MRTS}_C = \left|\frac{MP_L^C}{MP_K^C}\right| < \frac{w}{r}.$$

Similar discussion can be applied.



- Given TP_0, the lowest production cost could be implied by the isocost curve C_0. The corresponding input bundle is $A_0 = (L^*, K^*)$

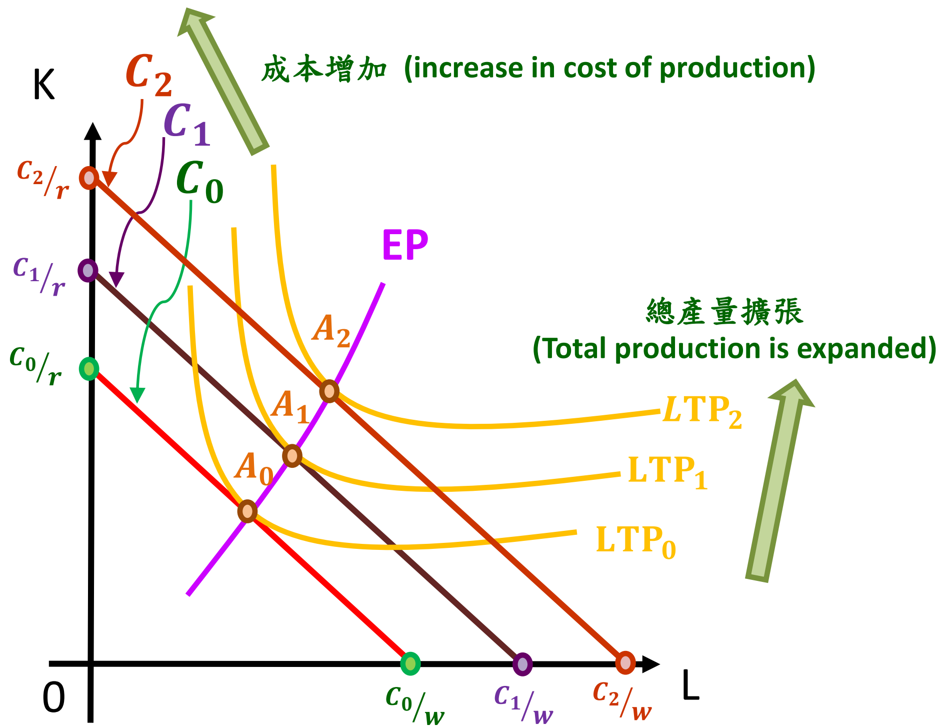
- Similarly, if the total product is increased to LTP_1 and LTP_2, the lowest costs of production are C_1 and C_2 and the corresponding inputs bundles are A_1 and A_2.

- Shift our focus on the relation between total product and total cost of production (LTC)

- Map the LTP and the corresponding lowest LTC to the LTP-LTC plane

- Connect all of the LTP and LTC, we obtain the firm's long-run cost curve (LTC)

- Notice that we let $\mathrm{LTC} = 0$ as $\mathrm{LTP} = 0$



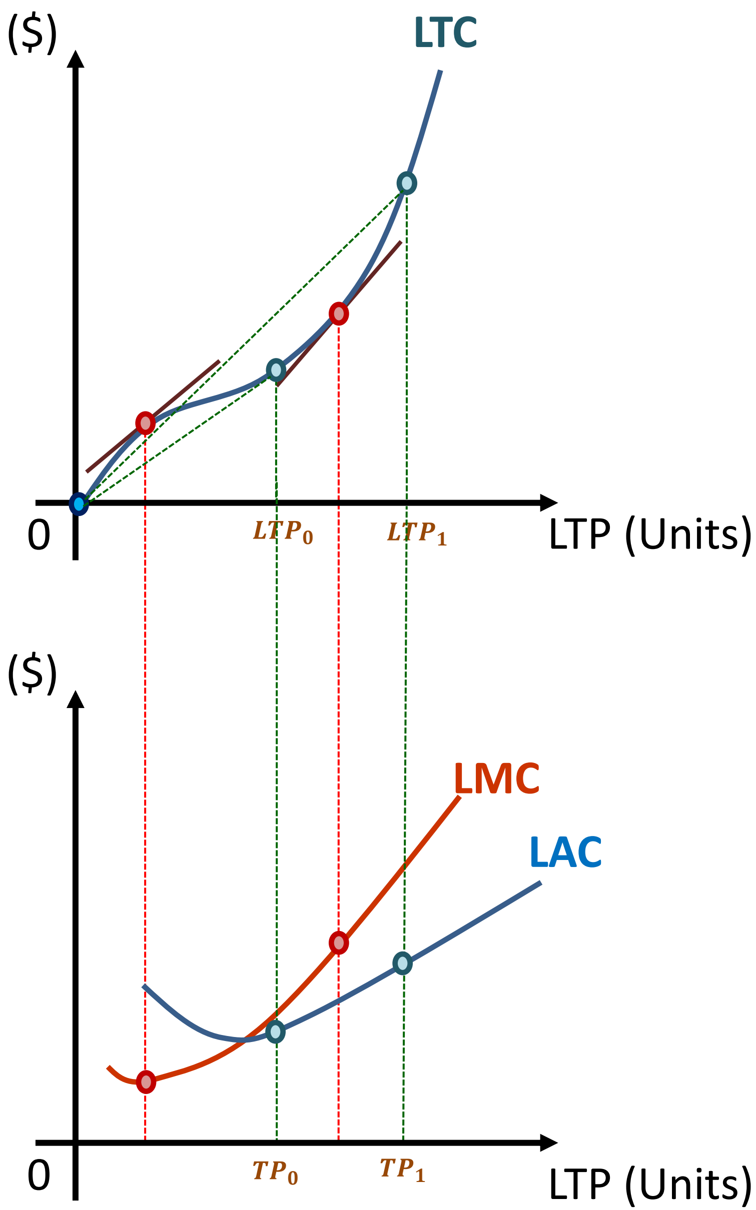
- 隨著總產量不斷的擴張，最低成本也不斷在增加，從 A_0、A_1、A_2所對應的LTP與C可以看出來

(C increases with LTP. This truth can be observed through C_0, C_1, C_2 implied by A_0, A_1, A_2)

- 將A_0、A_1、A_2 三點所連成線，稱為擴張曲線 (Expanding Path，EP)，描述了隨著總產量的增加，最低成本下的生產要素組合如何改變

(Connecting A_0, A_1, A_2 we get Expanding path (EP). The path tell us how optimal inputs bundle changes as TP increases)

- **The definition of Long-run Average Cost (LAC) and Long-run Marginal Cost (LMC)**



- **Long-run average cost (LAC) :**

  The long-run cost divided by total output

$$\mathbf{LAC} = \frac{\mathbf{LTC}}{\mathbf{LTP}}$$

It is the slope of the line connected by the point on the LTC curve and the origin

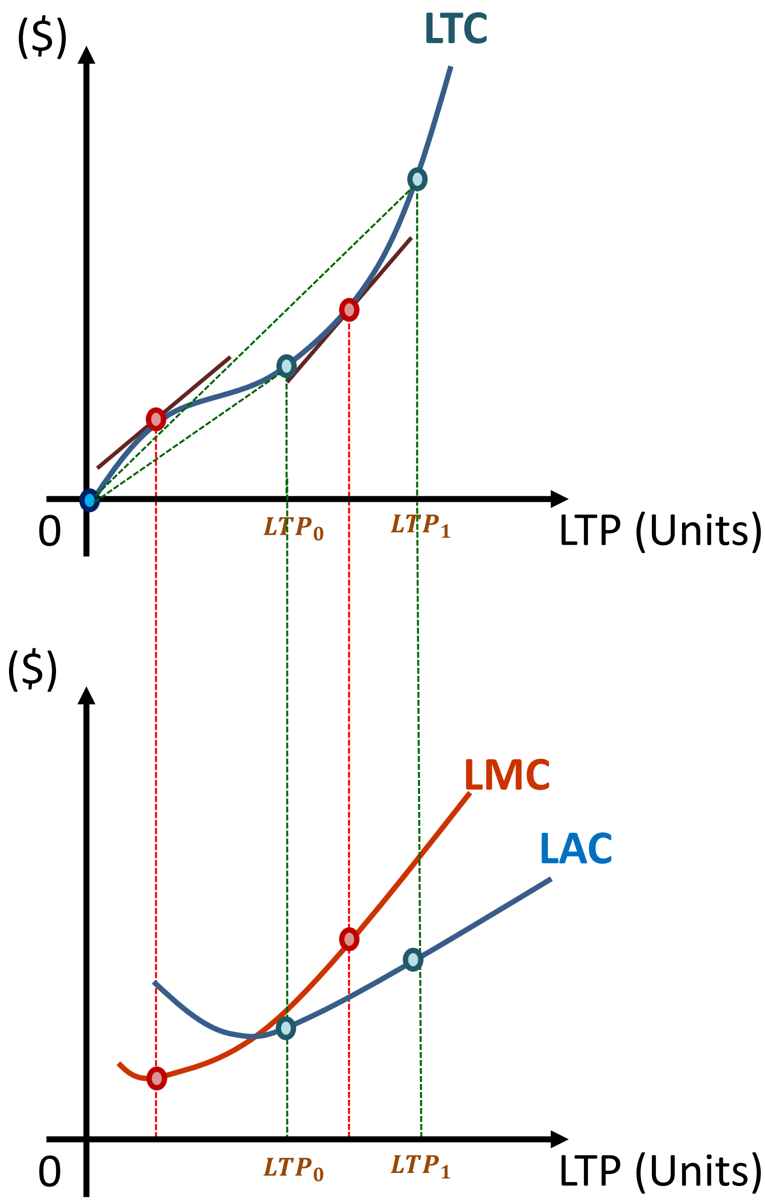
- **Long-run marginal cost (LMC) :**

  The increase in long-run cost that arises from an extra unit of production

$$\mathbf{LMC} = \frac{\mathbf{\Delta LTC}}{\mathbf{\Delta LTP}}$$

It is the tangent slope of the LTC

- **The relation between LAC and LMC**



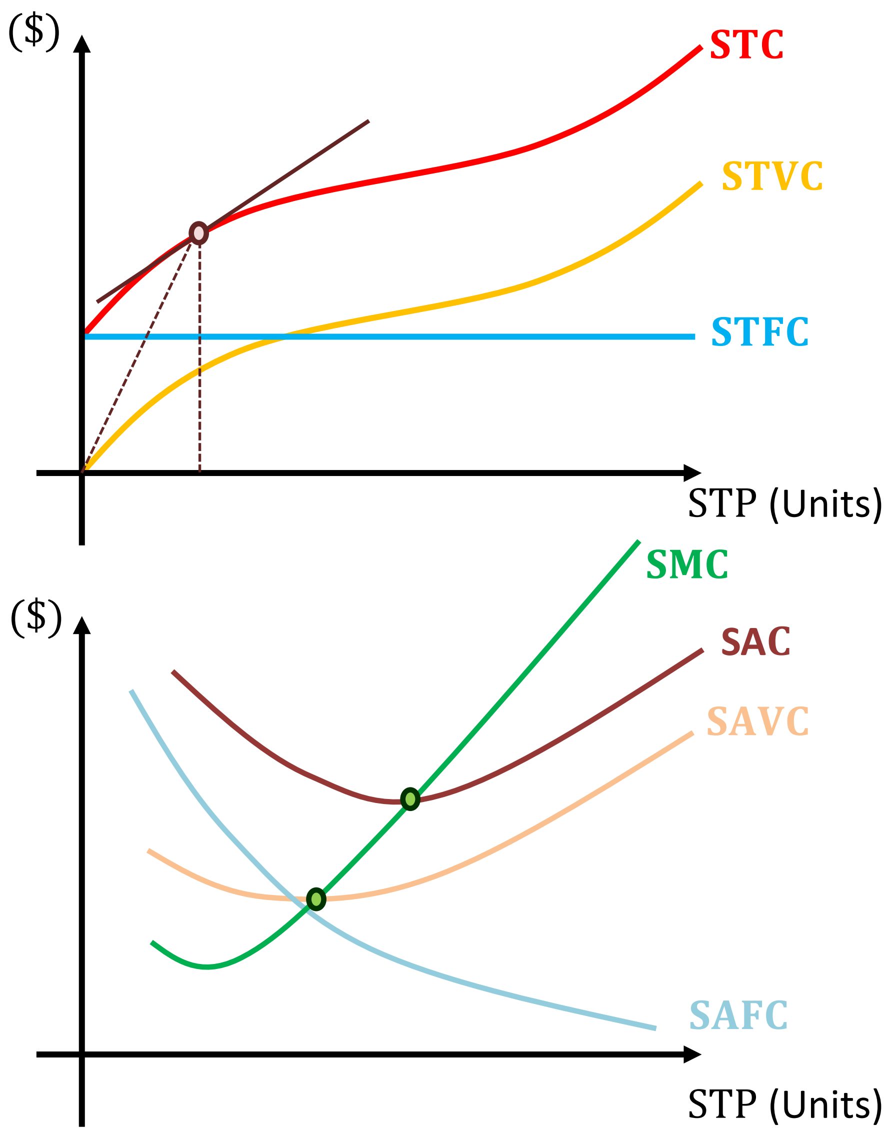
- **When LMC < LAC, LAC is decreasing with TP; when LMC > LAC, LAC is increasing with TP. Thus, when LMC = LAC, LAC is the minimum.**

- **Thus, the LMC curve will cross the lowest point of the LAC curve**

5-3.3: 長短期成本結構間的關係

(The relation between short- and long-run costs)

- **Recall: the relation between SAC and SMC**



- **The points on the SAC curve is the slope of the lines connected by the origin and the points on STC**

$$\mathrm{SAC} = \frac{\mathrm{STC}}{\mathrm{STP}}$$

- **The points on the SMC curve is the tangent slope of the STC or STVC curve**

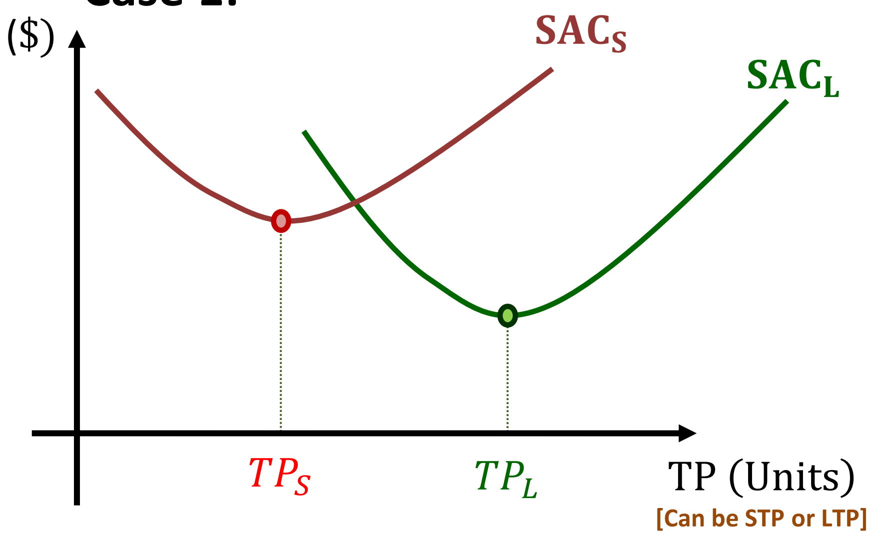
$$\mathrm{SMC} = \frac{\Delta \mathrm{STC}}{\Delta \mathrm{STP}} = \frac{\Delta \mathrm{STVC}}{\Delta \mathrm{STP}}$$

- **A SAC curve is U-shaped**

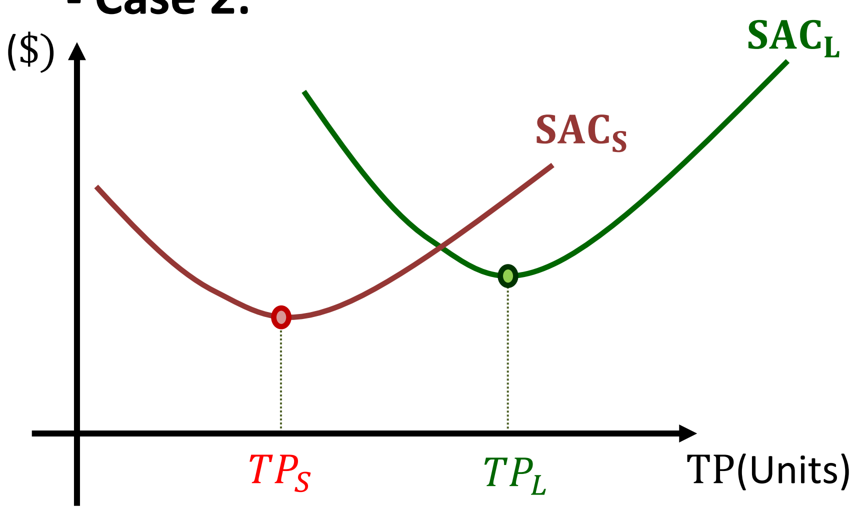
- **A SMC curve will cross the lowest point of its SAC、SAVC curve**

- **The relation between firm size (or factory size) and SAC**

- Case 1:



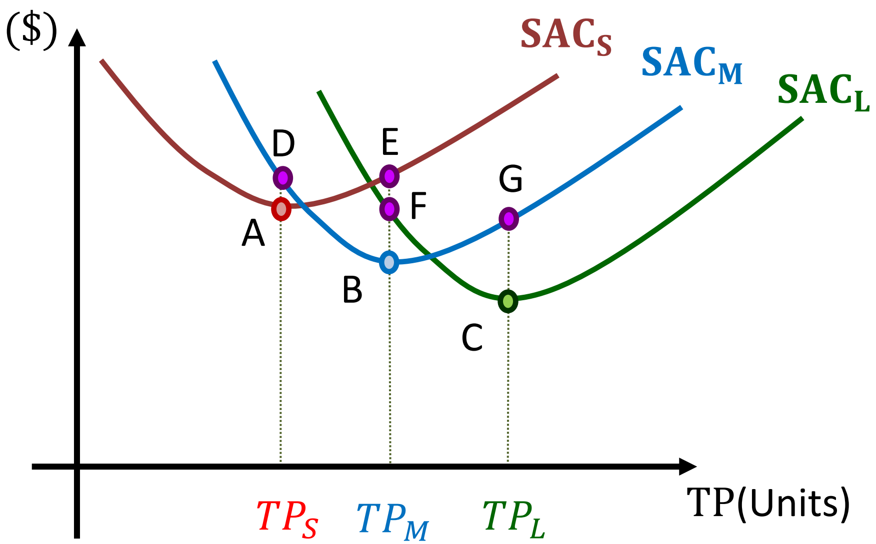
- Case 2:



Small firm and its TP with the lowest SAC

Large firm and its TP with the lowest SAC

- **The more the use of production inputs, the greater the firm size**
- **Given the lowest production cost used, the larger firm will have greater TP because it consumes more inputs**
- **Notice that SAC curves are U-shaped.**

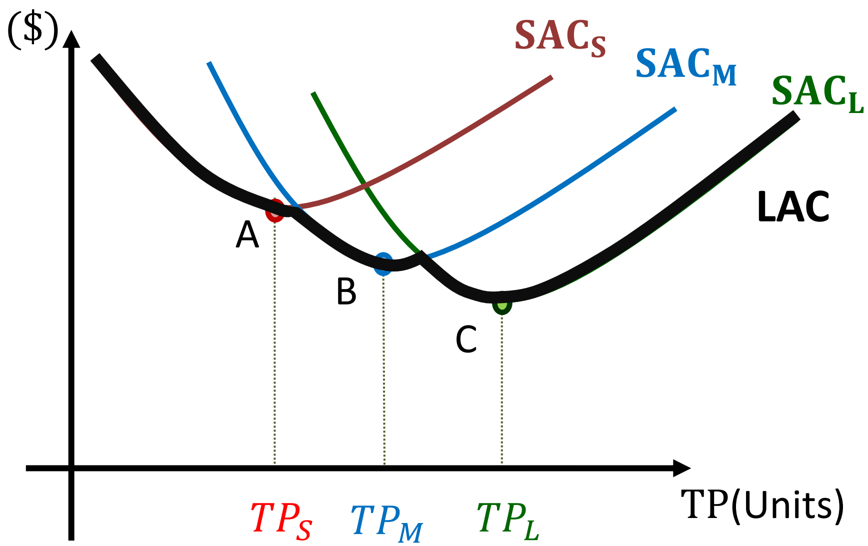


- **Change the firm size: it means the firm can change the quantity of inputs. In the short run, it can adjust the quantity of variable factors; in the long run, it can also adjust the quantity of fixed factors**

- **To reach TP_S, the cost implied by A is lower than that implied by D. It means that to reach TP_S, a rational producer will choose small size firm**

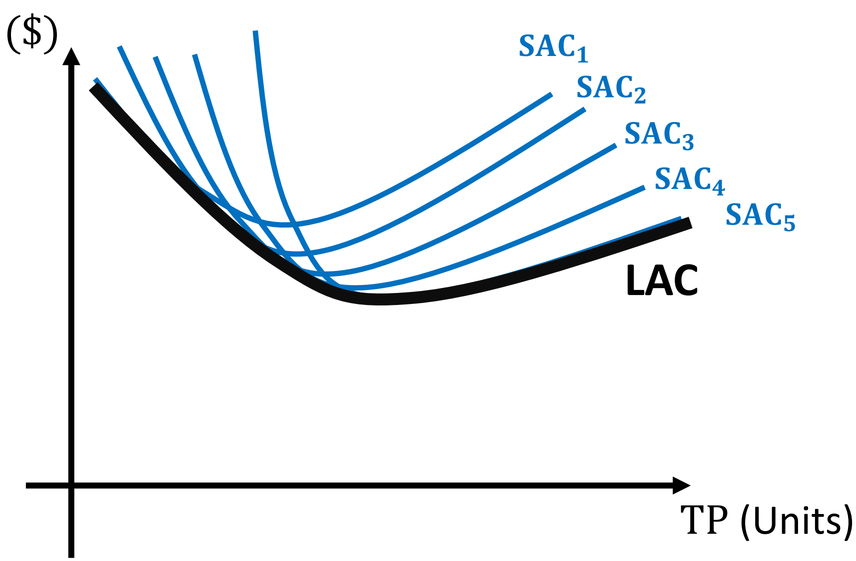
- **In the same token, to reach TP_M, the cost implied by B is lower than those implied by E and F. Thus, a rational producer will choose middle size firm.**

- **The relation between SAC and LAC**



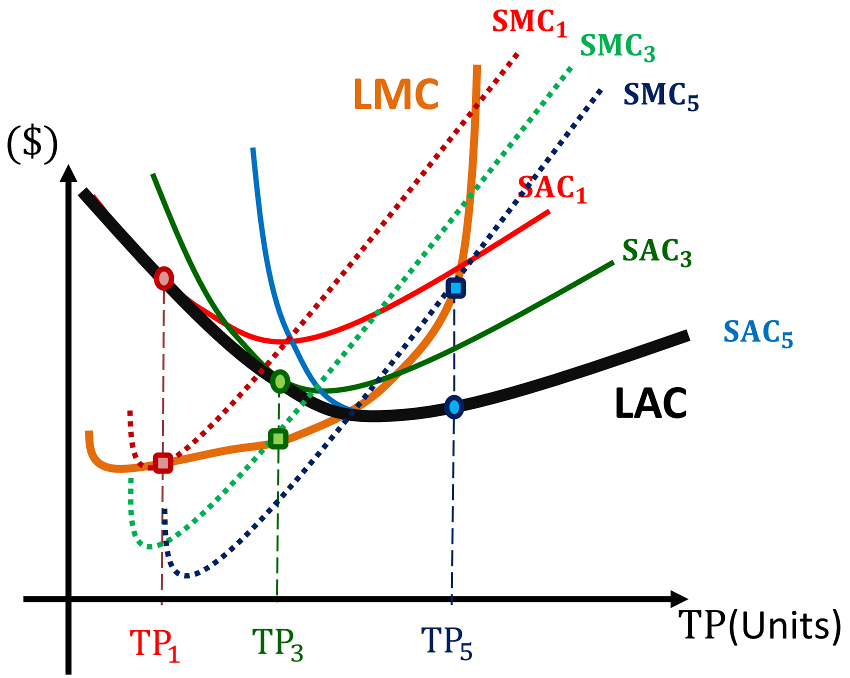
- **Thus in the long run, rational producers will choose the firm size having the LAC illustrated as the black curve.**

- **We say that a producer's LAC curve is the envelop curve comprised by SAC curves. The envelop curve is the lowest SAC under different STPs**



- **A LAC curve will become smooth once we have lots of SAC curves.**

- **The relation between SMC and LMC**



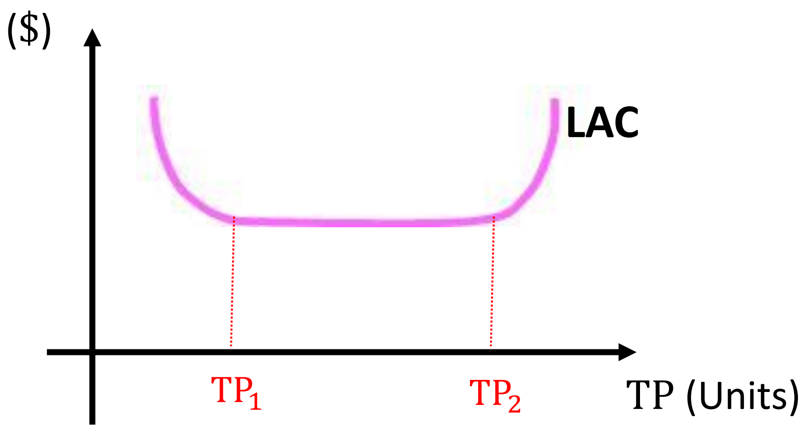
- **If a firm wants to reach $\mathbf{TP_1}$, it will choose size 1 having $\mathbf{SAC_1}$. It will choose size 3 having $\mathbf{SAC_3}$ if it wants to reach $\mathbf{TP_3}$. It will choose size 5 having $\mathbf{SAC_5}$ if it wants to reach $\mathbf{TP_5}$.**

- **The envelop curve of these SACs is LAC curve.**

- **Each of these SACs has a corresponding SMC that crosses the its lowest point**

- **Connecting all of these SMC, we obtain a LMC that crosses the lowest point of LAC curve**

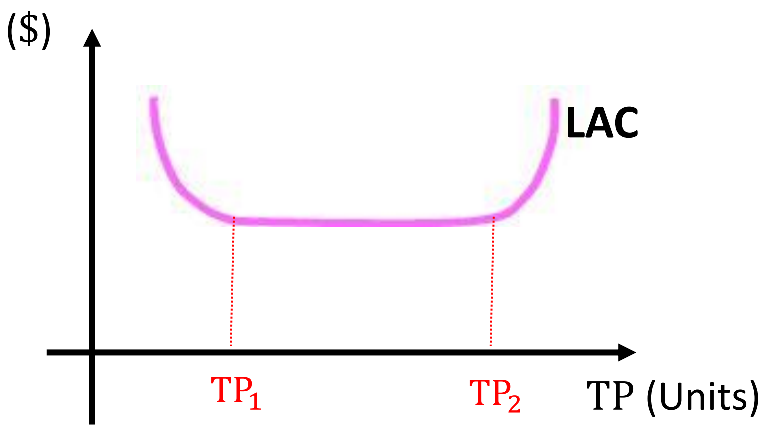
- **Three scenarios on the LAC curve**
  - **Generally speaking, a LAC curve is U-shaped according to the discussion in previous slides**
  - **Thus, the LAC could be increasing, constant, and decreasing when TP increases**



(1) Economies of scale: LAC falls as the quantity of TP increases. It is the case when TP is low (TP < TP_1)

(2) Diseconomies of scale: LAC rises as the quantity of TP increases. It is the case when TP is high (TP > TP_2)

(3) Constant returns to scale: LAC stays the same as the quantity of TP changes.



- **In the early stage of the firm, the firm's TP is still low. Its production will have economies of scale by increasing specialization among workers or by making massive purchase.**

- **When the firm's organization become larger and more complicated, its production will have diseconomies of scale because coordination problems are increasing.**

- **If the firm raises its number of branches at the same cost per branch, it has constant returns to scale. However, if its number of branches keeps growing, it finally has diseconomies of scale.**